

Homework8

Kunaal Raghav

Part 1

```
library(ISLR2)
```

Warning: package 'ISLR2' was built under R version 4.5.2

```
data("College")
my.college <- College[-484, ] # remove an extreme case.
#my.college <- College[College$Apps <=16000, ] # remove several extreme case.
train.pct <- 0.78
set.seed(2025)
Z <- sample(nrow(my.college), floor(train.pct*nrow(my.college)))
college.data <- my.college[Z, ]
holdout.data <- my.college[-Z, ]
```

The Full model is below

```
college.lmF <- lm(Apps ~ ., data = college.data)
college.lmF
```

Call:

```
lm(formula = Apps ~ ., data = college.data)
```

Coefficients:

(Intercept)	PrivateYes	Accept	Enroll	Top10perc	Top25perc
-863.89173	-491.58504	1.30000	-0.25815	48.03415	-14.56869
F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal
0.06908	0.02739	-0.04816	0.16270	0.23095	0.03091
PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad.Rate
-7.71449	-4.18016	14.93323	-5.28327	0.08351	9.68531

Step wise Selection Model

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:ISLR2':

Boston

```
college.lmAIC <- stepAIC(college.lmF, direction = "both", trace = FALSE)
summary(college.lmAIC)
```

Call:

```
lm(formula = Apps ~ Private + Accept + Top10perc + Top25perc +
    F.Undergrad + Outstate + Room.Board + PhD + perc.alumni +
    Expend + Grad.Rate, data = college.data)
```

Residuals:

Min	1Q	Median	3Q	Max
-3115.7	-397.1	-67.3	321.7	6775.3

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-538.43206	289.17439	-1.862	0.063103 .
PrivateYes	-505.38691	152.52320	-3.314	0.000977 ***
Accept	1.25473	0.04845	25.897	< 2e-16 ***
Top10perc	47.29382	5.87795	8.046	4.68e-15 ***
Top25perc	-14.36344	4.80592	-2.989	0.002918 **
F.Undergrad	0.04824	0.02398	2.011	0.044729 *
Outstate	-0.05181	0.02062	-2.513	0.012241 *
Room.Board	0.17743	0.05238	3.387	0.000753 ***
PhD	-10.26175	3.44977	-2.975	0.003053 **
perc.alumni	-7.28759	4.52714	-1.610	0.107983
Expend	0.07970	0.01241	6.424	2.73e-10 ***
Grad.Rate	9.18486	3.23754	2.837	0.004710 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1017 on 593 degrees of freedom
Multiple R-squared: 0.9192, Adjusted R-squared: 0.9177
F-statistic: 612.9 on 11 and 593 DF, p-value: < 2.2e-16

Best Subset Model

```
library(leaps)

best.sub <- regsubsets(
  Apps ~ .,
  data = college.data,
  nvmax = 17
)

best.sum <- summary(best.sub)

which.min(best.sum$bic)
```

[1] 8

```
college.lmBIC <- lm(
  Apps ~ Private + Accept + Top10perc + Top25perc +
    Outstate + Room.Board + PhD + Expend,
  data = college.data
)

summary(college.lmBIC)
```

Call:

```
lm(formula = Apps ~ Private + Accept + Top10perc + Top25perc +
    Outstate + Room.Board + PhD + Expend, data = college.data)
```

Residuals:

Min	1Q	Median	3Q	Max
-3151.5	-418.5	-40.2	295.4	6846.6

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-160.10863	265.98457	-0.602	0.547438
PrivateYes	-529.79215	148.56895	-3.566	0.000392 ***
Accept	1.35142	0.02413	56.013	< 2e-16 ***

Top10perc	49.31376	5.88163	8.384	3.69e-16	***
Top25perc	-14.07833	4.81038	-2.927	0.003557	**
Outstate	-0.05823	0.01947	-2.991	0.002897	**
Room.Board	0.19371	0.05182	3.738	0.000203	***
PhD	-10.04742	3.45634	-2.907	0.003785	**
Expend	0.07668	0.01244	6.164	1.31e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1024 on 596 degrees of freedom

Multiple R-squared: 0.9176, Adjusted R-squared: 0.9165

F-statistic: 829.2 on 8 and 596 DF, p-value: < 2.2e-16

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).